



ADAPTIVE TRAFFIC CONTROL SYSTEM

Intelligent Adaptive Traffic Control System

with Real-Time Pedestrian & Commuter Feedback

Team Members

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Critical Challenges

Problem Statement

01

Fixed Traffic Signal Timings

Static cycle durations fail to adapt to real-time traffic variations, causing unnecessary delays during low-traffic periods and congestion during peak hours.

02

Inefficient Traffic Density Handling

Traditional systems cannot dynamically allocate green time based on actual vehicle count, leading to imbalanced intersection throughput.

03

Unsafe Pedestrian Crossings

Lack of on-demand pedestrian signals creates dangerous situations where pedestrians cross against traffic or wait excessively.

04

No Real-Time Commuter Feedback

Drivers lack visibility into signal timing, causing anxiety, aggressive driving, and poor decision-making at intersections.

Increased Congestion & Fuel Wastage

Cumulative inefficiencies result in longer commute times, higher fuel consumption, and increased vehicular emissions.



Key Insight

Traditional traffic systems operate on **pre-programmed cycles**, while modern adaptive systems respond to **real-time demand**, improving efficiency by up to **30%**.



Project Goals

Objectives



01 Adaptive Signal Controller

Design and implement an intelligent traffic signal controller that dynamically adjusts signal timings based on real-time traffic conditions and pedestrian demand.

Dynamic Timing

Smart Logic

Real-Time



02 Traffic Density Sensing

Integrate ultrasonic sensor arrays to accurately detect and estimate vehicle density at each intersection approach for data-driven decision making.

Ultrasonic

Density Est.

Multi-Lane



03 Pedestrian Support

Provide on-demand pedestrian crossing functionality with push-button activation and priority-based scheduling integrated into the traffic control algorithm.

Push-Button

Priority

Safe Crossing



04 Real-Time Feedback Display

Implement countdown timer displays to provide commuters with clear visibility into signal timing, reducing uncertainty and improving traffic discipline.

7-Segment

Countdown

Visibility



05 Distributed Embedded Control

Demonstrate a complete distributed embedded system using dual microcontrollers (STM32 & Vega) with UART communication for real-time coordination.

STM32

Vega

UART



Why This Matters

Motivation



Growing Urban Traffic Congestion

Urban areas face **increasing traffic density** due to rapid urbanization. Traditional systems cannot scale, causing average commute times to increase by **15-20%** annually in major cities.



Need for Safer Pedestrian Infrastructure

Pedestrian fatalities at intersections account for **20% of all traffic deaths**. Intelligent crossing systems can reduce accidents by providing dedicated, on-demand crossing windows.



Demand for Smart City Solutions

Smart city initiatives worldwide require **intelligent transportation systems** as core infrastructure. Adaptive traffic control is a foundational technology for future urban mobility.



Real-World Embedded Systems Application

This project demonstrates practical application of **microprocessor interfacing, sensor integration, and real-time control algorithms**—essential skills for embedded systems engineers.

Improved Commuter Awareness



Real-time countdown displays reduce driver anxiety and **promote traffic discipline**, leading to smoother traffic flow and reduced aggressive driving behaviour.



Design Overview



Four-Lane Intersection Model

Complete intersection simulation with **North, South, East, and West** approaches. Each lane equipped with independent sensing and signal control for realistic traffic management.



Sensor-Based Detection

Ultrasonic sensors at each approach continuously monitor vehicle presence and estimate traffic density for intelligent decision-making.



Pedestrian Push-Button

Dedicated **push-button interface** at each crossing allows pedestrians to request crossing, which is queued and prioritized by the control algorithm.



Adaptive Timing Allocation

Intelligent algorithm calculates **optimal green time** based on traffic density, pedestrian requests, and historical patterns for maximum throughput.

System Block Diagram



SENSORS

Ultrasonic + Buttons



CONTROLLERS

STM32 + Vega Board



Signals

LED Indicators



Displays

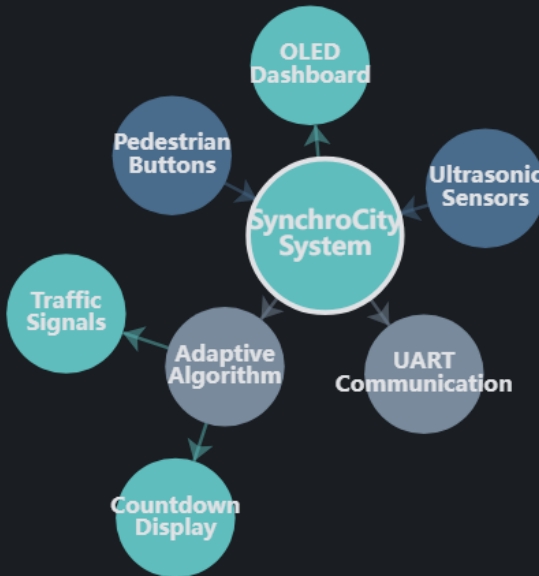
7-Seg + OLED



Key Feature: Distributed architecture with dedicated sensor processing and centralized logic control enables real-time responsiveness.



Proposed Solution



Ultrasonic Density Estimation

Sensors measure distance to vehicles to estimate lane occupancy and traffic density with **$\pm 5\text{cm}$ accuracy**.



Weighted Adaptive Scheduling

Algorithm assigns green time based on **weighted traffic density** across all lanes, optimizing intersection throughput.



Pedestrian Priority Integration

Push-button requests are **queued and prioritized** based on wait time, ensuring safe crossing opportunities.



Real-Time Countdown Timer

7-segment displays show remaining green/red time, improving commuter awareness and traffic discipline.



OLED Monitoring Dashboard

Real-time system status display showing **lane densities, signal states, and pedestrian queue** for debugging and monitoring.



Hardware Implementation

Embedded Design Architecture



Sensor Controller

STM32F439ZI Board



Sensor Data Acquisition

Reads ultrasonic sensor values via GPIO and ADC interfaces at **10Hz** sampling rate



Traffic Density Estimation

Processes sensor data to calculate lane occupancy and vehicle count estimation



Pedestrian Input Processing

Detects and debounces push-button presses, maintains pedestrian request queue



UART Data Transmission

Transmits density data and pedestrian requests to Logic Controller at **115200 baud**



UART

115200 baud



Logic Controller

Vega Board



Traffic Scheduling Algorithm

Implements weighted adaptive scheduling with pedestrian priority integration



Signal LED Control

Drives RGB LEDs for traffic signals (Red/Yellow/Green) via GPIO pins



Countdown Timer Display

Controls 7-segment displays to show remaining signal time for each lane



OLED Dashboard Management

Updates OLED display with real-time system status and debugging information



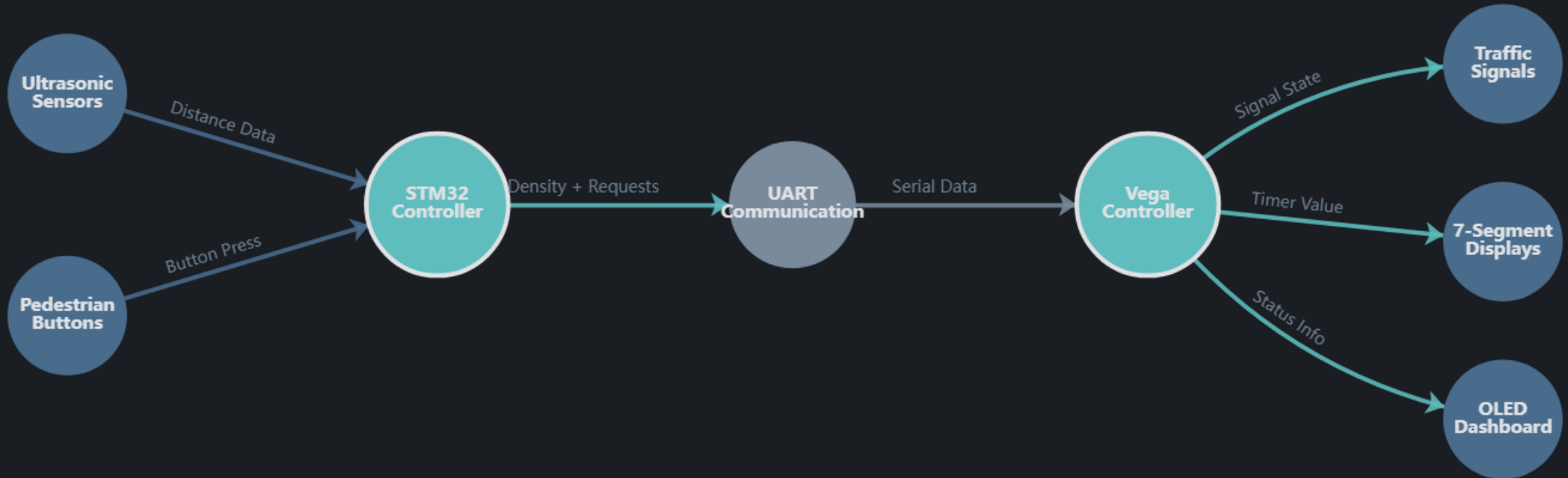
Distributed Processing: Sensor and logic tasks separated for efficiency



Real-Time Communication: UART ensures synchronized operation



Component Interaction & Data Flow





Project Deliverables

Expected Outcomes



Materialistic Outcomes



Functional Prototype

A fully operational **4-lane intersection model** demonstrating adaptive signal control with real-time sensor integration and pedestrian management.



Embedded System Design

Complete hardware-software integration using **STM32 and Vega boards** with documented circuit diagrams and communication protocols.



Performance Metrics

Quantified improvements in **traffic throughput, wait times, and pedestrian safety** compared to fixed-timing systems.



Learning Outcomes



Embedded Systems Expertise

Hands-on experience with **microcontroller programming, sensor interfacing, and real-time control systems** using industry-standard development tools.



Algorithm Development

Practical understanding of **scheduling algorithms, priority management, and optimization techniques** for resource-constrained environments.



Team Collaboration

Experience in **project management, technical documentation, and collaborative problem-solving** in a multi-disciplinary team environment.



Demonstrable Prototype



Technical Documentation



Performance Analysis



Thank You

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Course

ES333 Microprocessors and Embedded Systems

Project Type

Academic Engineering Project



Embedded Systems



Smart Traffic



Adaptive Control